

DYESS AFB, TX, USA

WMO: 720965

Lat: 32.433N

Lon: 99.850W

Elev: 545

StdP: 94.95

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.2	(c) -5.0	(d) -15.2	(e) 1.1	(f) -0.1	(g) -12.8	(h) 1.3	(i) 0.9	(j) 13.9	(k) 11.3	(l) 12.5	(m) 11.4	(n) 3.8	(o) 0	(p) 0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	39.1	22.2	37.8	22.1	36.7	22.1	24.8	33.7	24.1	32.9	23.7	32.3	5.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.4	27.4	22.1	18.0	27.2	21.4	17.1	27.1	78.1	33.6	75.6	32.8	73.5	32.4	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.1	11.0	9.7	-10.8	40.9	2.3	1.4	-12.4	41.9	-13.8	42.7	-15.1	43.4	-16.8	44.4
			-11.5	25.9	2.0	1.0	-12.9	26.7	-14.1	27.2	-15.2	27.8	-16.7	28.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.7	7.2	9.7	13.9	18.4	23.3	27.7	29.3	28.8	25.0	19.1	12.7	8.2
	DBStd	9.22	5.95	6.08	6.04	4.94	4.63	2.96	2.44	2.66	3.84	5.10	5.42	5.75
	HDD10.0	377	127	73	32	3	0	0	0	0	0	4	35	103
	HDD18.3	1379	347	248	161	59	13	0	0	0	3	53	179	316
	CDD10.0	3534	39	63	152	256	412	531	598	584	449	286	116	48
	CDD18.3	1493	1	5	22	61	166	281	340	326	202	77	10	2
	CDH23.3	18258	24	82	266	706	1878	3523	4497	4130	2246	774	101	31
	CDH26.7	9652	3	23	77	271	914	1935	2623	2378	1130	276	18	3
(14) Wind	WSAvg	4.7	4.6	4.7	5.3	5.6	5.3	5.2	4.3	3.8	3.8	4.4	4.5	4.4
(15) Precipitation	PrecAvg	649	28	30	41	55	87	78	60	59	71	62	39	27
	PrecMax	899	81	119	99	141	196	195	243	187	191	150	116	174
	PrecMin	309	0	0	0	5	19	2	3	0	6	8	0	0
	PrecStd	166	24	31	28	35	43	55	59	50	52	41	36	34
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	30.1	31.6	34.8	38.9	40.9	40.4	40.3	38.6	34.1	28.9	26.7
		MCWB	14.1	14.6	16.4	17.1	20.0	22.6	22.2	22.5	21.5	20.2	16.6	13.8
	2%	DB	22.8	25.8	28.9	32.2	36.3	38.4	38.9	38.8	36.2	31.9	26.1	23.0
		MCWB	12.3	14.0	16.1	17.1	20.6	22.4	22.5	22.3	21.7	19.8	16.2	13.2
	5%	DB	20.0	22.7	26.5	29.9	34.0	36.5	37.6	37.4	34.5	29.8	23.8	20.4
		MCWB	10.9	13.5	15.5	17.3	20.4	22.3	22.3	22.3	21.3	19.5	15.3	12.1
	10%	DB	17.4	19.9	23.9	27.6	31.8	34.8	36.3	36.0	32.8	27.7	21.7	17.8
		MCWB	10.0	12.3	14.6	16.8	20.3	22.3	22.3	22.3	21.2	18.3	14.2	11.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	18.3	20.7	21.7	24.0	25.7	25.6	25.1	24.2	23.2	20.3	17.9
		MCDB	21.5	22.8	24.5	29.5	32.1	35.2	35.0	33.7	31.9	28.6	25.6	21.5
	2%	WB	14.1	16.5	19.1	20.5	23.0	24.8	24.7	24.4	23.5	22.0	18.4	15.5
		MCDB	19.4	21.5	24.6	27.2	31.3	33.7	33.8	33.2	31.2	27.9	22.4	19.9
	5%	WB	12.5	14.8	17.7	19.5	22.3	24.0	24.0	23.9	22.8	21.1	17.1	13.5
		MCDB	17.8	20.6	22.7	25.5	30.3	32.6	33.0	32.7	30.5	27.0	21.7	18.6
	10%	WB	10.7	12.9	16.3	18.4	21.5	23.4	23.5	23.4	22.2	20.1	15.7	11.6
		MCDB	17.1	19.5	22.0	24.1	29.2	31.7	32.4	32.1	29.5	25.9	20.5	17.3
(35) Mean Daily Temperature Range	MDBR		14.2	13.7	14.2	14.5	13.2	12.6	12.8	12.6	12.6	13.6	13.6	13.3
	5% DB	MCDBR	17.4	17.7	16.7	16.7	15.8	14.6	14.3	14.2	14.5	14.8	15.0	17.1
		MCWBR	9.4	9.5	8.1	7.7	5.9	4.6	4.3	4.3	4.8	6.7	7.9	9.7
	5% WB	MCDBR	15.2	15.5	13.3	13.5	13.3	12.8	12.6	12.6	11.6	12.1	12.3	13.7
		MCWBR	9.7	9.7	7.6	7.5	5.7	4.7	4.3	4.1	4.4	6.2	8.0	9.3
(40) Clear-Sky Solar Irradiance	taub		0.289	0.303	0.324	0.367	0.389	0.406	0.416	0.404	0.374	0.351	0.318	0.297
	taud		2.544	2.507	2.466	2.355	2.364	2.346	2.379	2.397	2.419	2.465	2.489	2.526
	Ebn at Noon		943	959	957	922	899	878	868	876	892	890	896	911
	Edh at Noon		82	94	106	124	124	126	121	117	110	98	87	80
(44) All-Sky Solar Radiation	RadAvg		3.21	3.88	5.09	6.19	6.50	7.12	6.95	6.36	5.32	4.27	3.38	2.82
	RadStd		0.34	0.52	0.35	0.47	0.63	0.53	0.45	0.43	0.40	0.62	0.40	0.32

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (16 neighbors)		+0.33	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	+84

Nomenclature: See separate page